Next-Day Stock Status Prediction in Fresh Retail

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Problem Statement & Challenges

The Fresh Retail Dilemma

Fresh retail operations face a constant battle between **overstocking** (leading to spoilage and massive food waste) and **understocking** (leading to lost revenue, stockouts, and customer churn). Accurate, intraday prediction of out-of-stock (OOS) events is critical for dynamic shelf replenishment.

Core Modeling Challenges

- **Censored Demand (Stockout-Induced Bias)**: When an item is out of stock, sales record as zero. Traditional forecasting models misinterpret this as "zero demand", leading to a negative feedback loop of systematic under-ordering.
- **Severe Class Imbalance**: True stockout hours are rare anomalies (often $< 5\%$ of total operational hours), making it extremely difficult for standard ML loss functions to learn the minority class without collapsing.
- **High-Velocity Intraday Dynamics**: Unlike durable goods, fresh produce sales exhibit extreme intraday volatility driven by time-of-day foot traffic, flash sales, and perishable shelf-life windows.
- **Hierarchical Spatial-Temporal Variance**: Demand shocks are not uniform; they vary heavily across different city zones, specific store demographics, and individual SKU categories.

Literature Review & Related Work

General Stockout Prediction Research

- **Classical ML Ensembles** (Andaur et al., 2021): Showcased Random Forest and voting ensembles as robust benchmarks for retail stockout prediction on tabular data.
- **Deep Learning Architectures** (Giaconia & Chamas, 2023): Utilized massive CNN, Self-Attention, and Temporal Convolutional Networks (TCN) ($>$ 500k parameters) for intraday stockout probability.
- **Large-Scale Imbalance Handling** (Liu et al., 2025): Benchmarked XGBoost and SMOTE strategies, proving that near-term demand indicators dominate long-term forecasts for stockout events.

Research Utilizing the FreshRetail Dataset

- **Hierarchical Forecasting**: *"A Coupled Hierarchical Architecture for Multi-Granularity Demand Forecasting"* uses this exact dataset to tackle the complex spatial/product hierarchy (City $\rightarrow$ Store $\rightarrow$ SKU).
- **Censored Demand Bias**: *"When Less Data Is Better: Eliminating Stockout-Induced Bias in Choice Estimation"* explicitly models how stockouts artificially deflate true demand signals in FreshRetail environments.

Dataset Overview & Processing Strategy

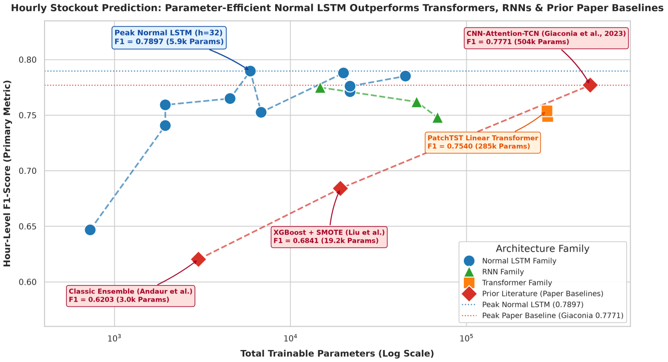
FreshRetailNet-50K Scale & Scope

- **Volume**: 50,000 store-product pairs across 898 stores.
- **Timeframe**: 90-day series of highly granular hourly data (Chronological Split: First 75 days for Training, Final 15 days for Validation).

Why We Selected This Dataset

- **Pioneering Benchmark**: First large-scale public dataset designed for **censored demand estimation** in fresh retail (where out-of-stock items hide true demand).
- **Targeted Selection**: We isolated the **Top 15 SKUs** using a custom ranking formula to focus purely on high-impact, stockout-prone goods and eliminate noise from dormant items.

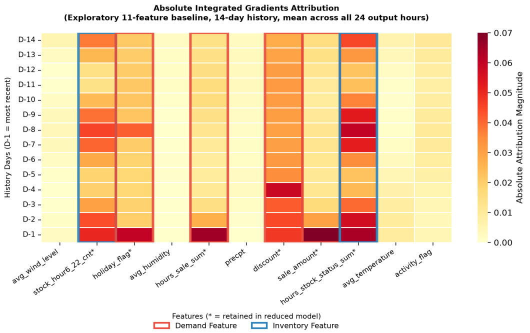
Model Family Comparison: Simple LSTMs Beat Heavy Transformers



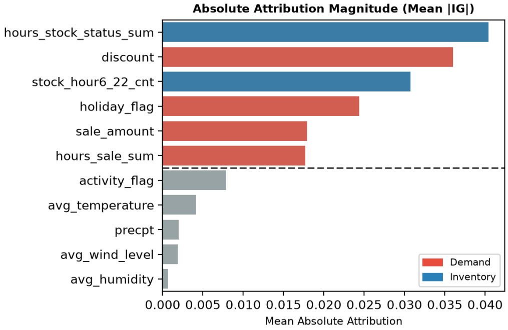
Key Empirical Findings

- **Parameter Efficiency**: Normal LSTMs (5.9k–44.9k params) reach **0.7852–0.7897 F1-Score**, matching or beating heavy Transformers.
- **Transformer Overparameterization**: PatchTST (285k params) & TFT (288k params) peak at **0.7540–0.7685 F1-Score**.
- **Why LSTMs Win**: Sequential state memory h_t naturally aligns with continuous hourly depletion better than permutation-invariant self-attention.

ML Explainability & Baseline Limitations



Attribution heatmap: 11-feature baseline (D-1 = most recent)



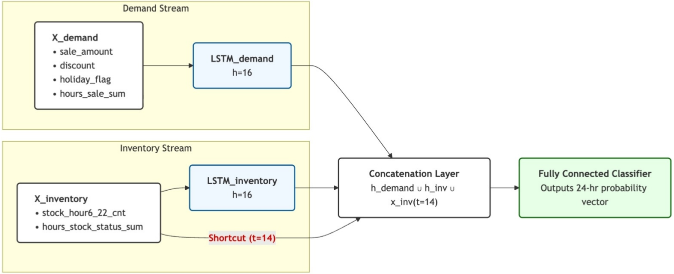
Feature importance: top 6 (highlighted) outrank remaining 5

Motivation for Dual-Stream Architecture

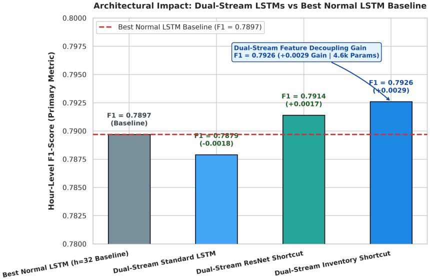
- **F1 Plateau:** Baseline LSTM peaks at ∼0.789 F1; simply scaling parameters yields no gain.
- **Interference:** Sharing a single hidden state h_t creates potential for demand/inventory interference.
- **IG Analysis:** Absolute attribution reveals distinct temporal behaviors for demand vs. inventory.

These findings motivate processing both streams independently before a task-specific fusion layer.

Our Fix: Dual-Stream Inventory Shortcut LSTM



Dual-Stream: Demand LSTM + Inventory LSTM + Day-14 Shortcut



Architectural Impact: Dual Stream LSTM

Key Results

- **Inventory Shortcut**: F1 **0.7926**, MAE **3.97 hrs** at just 4,600 params.
- Decoupled streams consistently beat all baseline LSTM sizes.

Exploring Linear Decomposition: DLinear Models

Why Explore Linear Decomposition?

DLinear (Zeng et al., 2023) demonstrated that single-layer linear models decomposing time series into Trend and Seasonal components can outperform complex deep networks.

DLinear Series Decomposition

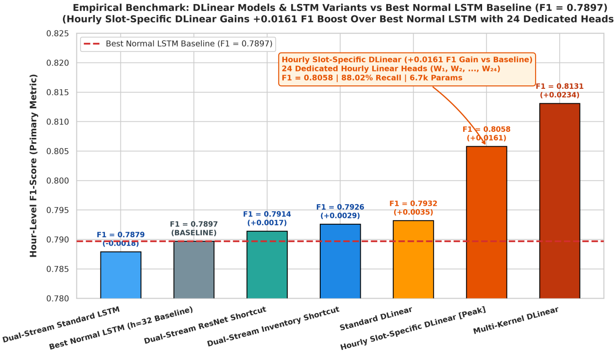
$$X_{\text{trend}} = \text{AvgPool}(X), \quad X_{\text{seasonal}} = X - X_{\text{trend}}$$

$$\mathbf{Y}_{\text{pred}} = W_{\text{trend}} X_{\text{trend}} + W_{\text{seasonal}} X_{\text{seasonal}}$$

Standard DLinear Performance

- **Parameters**: 6,768 parameters.
- **Validation F1-Score**: **0.7932**
  - Outperforms Normal LSTMs (**0.7897**)
  - Outperforms Enhanced LSTMs (**0.7926**)
  - Outperforms Transformers (**0.7540**)
- **Stockout Recall**: **87.44%**.

Architectural Innovation 2: Linear Changes in DLinear



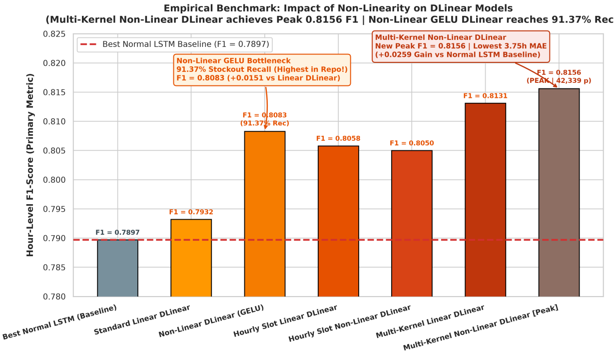
Hourly Slot-Specific Linear DLinear

- Uses **24 dedicated linear heads** $(W_1, \ldots, W_{24})$ for each operational hour.
- Resolves inter-hour gradient conflict between morning and evening rush hours.
- Boosts F1 to **0.8058** (+0.0161 over LSTM).

Multi-Kernel Linear DLinear

- Multi-scale moving average trend pooling ($k \in \{3, 5, 7\}$).
- Reaches **0.8131 F1-Score** with pure linear decomposition (13.5k params).

Architectural Innovation 3: Introducing Non-Linearity into DLinear



Non-Linear Activation Bottleneck

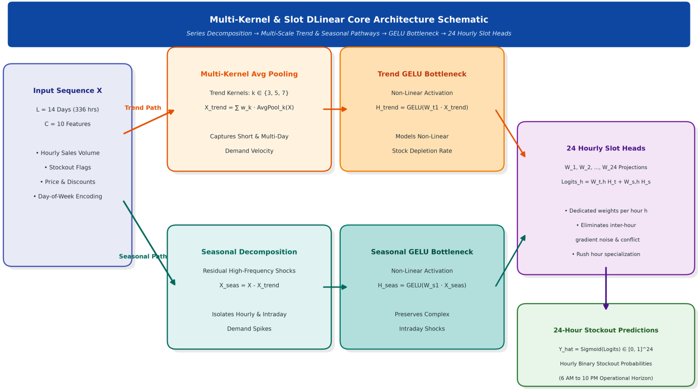
Adds GELU activation bottlenecks between decomposition and output projections:

$$\hat{Y} = W_{\text{trend},2}\text{GELU}(W_{\text{trend},1}X_{\text{trend}}) + \text{Seasonal}$$

Breakthrough Results

- **Non-Linear GELU DLinear**: Reaches **91.37% Stockout Recall** (Highest in Repo!) and **0.8083 F1**.
- **Multi-Kernel Non-Linear DLinear**: Reaches **0.8156 F1-Score** (New Peak F1 & lowest 3.75h MAE).

Architectural Schematic: Multi-Kernel & Slot DLinear



Core Mechanism

- **Trend Stream**: Multi-kernel moving average pooling ($k \in \{3, 5, 7\}$) + GELU bottleneck.
- **Seasonal Stream**: High-frequency residual stockout shocks ($X - X_{\text{trend}}$).
- **24 Slot Heads**: $W_1, \ldots, W_{24}$ dedicated projections resolve inter-hour gradient conflicts.

Master Empirical Validation Leaderboard

Rank	Architecture Name	Category	Params	F1-Score	Recall	MAE (h
1	**Multi-Kernel Non-Linear DLinear**	Non-Linear DLinear	42,339	**0.8156**	87.13%	**3.75**
2	**Multi-Kernel Linear DLinear**	Linear (Multi-Scale)	13,539	**0.8131**	85.22%	3.86
3	**Non-Linear GELU DLinear**	Non-Linear DLinear	21,168	**0.8083**	**91.37%**	4.20
4	Hourly Slot-Specific DLinear	Linear (24 Heads)	6,744	**0.8058**	88.02%	4.23
5	Standard Linear DLinear	Linear (Decomp)	6,768	**0.7932**	87.44%	4.28
6	Dual-Stream Inventory Shortcut	LSTM Family	4,600	**0.7926**	80.79%	3.96
Base	**Best Normal LSTM (h=32)**	Normal LSTM	5,912	**0.7897**	81.29%	4.31
7	Feature-Reduced Standard LSTM	Normal LSTM	19,992	0.7880	81.60%	4.58
8	Baseline Standard LSTM (h=96)	Normal LSTM	44,952	0.7852	80.40%	4.67
9	Vanilla Simple RNN (h=64)	RNN Family	14,744	0.7750	81.02%	4.23
10	PatchTST Patch Linear	Transformer	285,624	0.7540	81.90%	4.16

Conclusion & Operational Deployment Takeaways

Key Architectural Takeaways

1. **Non-Linearity Boosts DLinear**: GELU bottlenecks boost DLinear F1 to **0.8083** and Recall to a record **91.37%**.
2. **New All-Time Peak F1**: Multi-Kernel Non-Linear DLinear sets the highest F1-Score of **0.8156** and lowest MAE of **3.75 hours**.
3. **Feature Decoupling & Slot Heads**: Resolves inter-hour gradient noise and temporal cross-talk.

Deployment Recommendations

- **Primary Production Deployment**: **Multi-Kernel Non-Linear DLinear** (0.8156 F1, 3.75h MAE, 42.3k params).
- **Maximum Risk Mitigation**: **Non-Linear GELU DLinear** (91.37% Stockout Recall).
- **Future Scope**: Incorporate real-time intraday point-of-sale stream updates for dynamic threshold adjustment.